JEGOS

FORM FOUR PRE-NATIONAL EXAMINATION

036 INFORMATION AND COMPUTER STUDIES
MARKING SCHEME 2026

SECTION A (16 MARKS)

1.

(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)	(ix)	(x)
B	C	D	A	C	E	C	A	B	B

1@ = 10 marks

2.

LIST A	(i)	(ii)	(iii)	(iv)	(v)	(vi)
LIST B	C	B	A	D	E	F

1@ = 6 marks

SECTION B (54 MARKS)

3.

(a) Three Warning Signs of Malware Infection: (3 Marks)

- Sudden, unexplained degradation in system processing speeds and long boot times.
- Frequent, unauthorized pop-up windows, unfamiliar error dialogues, or files turning into shortcuts.
- Spontaneous loss of data files, unexpected changes to file extensions, or system files being deleted.

(b) Three Common Entry Vectors: (3 Marks)

- Connecting infected removable flash drives or external storage media.
- Downloading unverified files or attachments from malicious emails or websites.
- Operating on unpatched local networks with out-of-date security firewalls.

(c) Three Preventive Measures: (3 Marks)

- Install and regularly update robust, professional anti-virus and anti-malware software.
- Keep the operating system security patches and network firewalls fully up to date.
- Run routine security education for users regarding safe email management and avoiding pirated software downloads.

4.

(a) Three reasons for protecting information systems (3 marks)

Award 1 mark each, any three.

- Prevent unauthorized access.
- Protect confidentiality of information.
- Maintain integrity of data.
- Prevent financial loss.
- Ensure business continuity.

• Improve customer confidence.

(b) Three ICT-related crimes (3 marks)

Examples:

• Hacking
• Identity theft
• Phishing
• Software piracy
• Cyber fraud
• Malware distribution
• Cyberbullying
• Online financial fraud

(c) Security measures (3 marks)

• Install antivirus software.
• Use strong passwords.
• Encrypt sensitive data.
• Configure firewalls.
• Update software regularly.
• Apply user access control.
• Train users on cyber security.

5.

(a) Three Health Risks: (3 Marks)

• Carpal Tunnel Syndrome or Repetitive Strain Injury (RSI) in fingers and wrists.
• Computer Vision Syndrome (CVS) leading to dry eyes, blurred vision, and headaches.
• Chronic neck, shoulder, and lower back pain due to slouched sitting habits.

(b) Three Ergonomic Solutions: (6 Marks)

• Adjust chair height and backrests to support a neutral spine position, keeping feet flat on the floor.
• Position the computer monitor at eye level, about an arm's length away, to reduce neck strain and eye fatigue.
• Use ergonomic keyboards and wrist rests to keep wrists in a natural, straight position while typing.

6.

(a) Two Advantages of Electronic Approach: (3 marks)

Immediate Processing and Entry: Data is transmitted directly to a central database, eliminating manual paperwork entry errors and long delays.

Built-in Validation Rules: The software catches data configuration anomalies instantly at the point of entry, improving data accuracy.

(b) Three Field Challenges: (3 marks)

• Sudden system outages or battery depletion in areas without access to electricity.
• Poor or unstable mobile network connections, which prevent real-time data synchronization.
• Hardware vulnerabilities, such as screen cracks or liquid damage from harsh outdoor environments.

(c) Two Advantages of Cloud Computing (3 marks)

- Enables real-time data synchronization.
- Allows access to information from different locations.
- Reduces risk of data loss through cloud backup.
- Facilitates easier sharing of information.
- Reduces dependence on local storage.

7.

(a) Steps for Conditional Formatting: (3 Marks)

- Highlight the target cell range using the mouse cursor.
- Click the Home tab on the main toolbar, locate the Styles group, and open the Conditional Formatting dropdown menu.
- Select Highlight Cells Rules → Less Than..., type 50 in the value box, choose Red Fill, and click OK.

(b) Steps for Freezing the Top Row: (3 Marks)

- Open the active worksheet and click on the View tab on the main menu ribbon.
- Locate the Window operational group and click on the Freeze Panes dropdown button.
- Select Freeze Top Row from the options to lock the header row in place.

(c) Steps for Protecting a Worksheet: (3 Marks)

- Click on the Review tab on the main toolbar ribbon.
- Locate the Protect group and click the Protect Sheet command button.
- Enter a secure password in the dialogue box, check the allowed user actions, click OK, re-enter the password to confirm, and save the file.

8.

(a) Storage capacity and processor speed (2 marks)

- Hard disk storage capacity: 1 TB
- Processor speed: 3.2 GHz

(b) Functions of Hard Disk (3 marks)

Award 1.5 marks each, any two.

- Permanent storage of programs.
- Stores user files.
- Stores operating system.
- Stores application software.
- Stores backups.

(c) Functions of RAM (4 marks)

- Temporarily stores data being processed.
- Temporarily stores running programs.
- Supplies CPU with data quickly.
- Holds intermediate processing results.

SECTION C (30 MARKS)

9.

Introduction (1.5 Marks): Define multimedia technology as the integration of multiple media elements—such as text, graphics, audio, animation, and video—processed digitally. State that its

growth serves as an engine for economic development and modern communications in Tanzania.

Main Body (12 Marks – 3 Marks per well-evaluated sector):

Education Sector: Interactive animations and digital video lectures make complex STEM subjects easier to understand in schools. E-learning platforms expand access to educational resources beyond physical classrooms.

Marketing and Business: Commercial firms use engaging digital video ads and graphics on social media to build brand visibility and boost sales, driving digital business growth.

Entertainment and Creative Arts: The local film and music industries rely on advanced digital audio-video editing tools to produce high-quality media, creating jobs for youth and boosting national entertainment revenue.

Public Health and Civic Campaigns: Government bodies deploy multimedia infographics and radio/television animations to share critical health information (e.g., vaccination drives or disease prevention), improving public awareness and safety.

Conclusion (1.5 Marks): Summarize that multimedia technology has transformed traditional communication channels. Conclude that continued investment in digital skills training and internet infrastructure is essential to maximize its socio-economic benefits.

10.

Introduction (1.5 Marks): Define an operating system (OS) as the core system software that manages computer hardware resources and provides common services for application programs. State that choosing the right OS requires balancing hardware limits with institutional needs.

Main Body (12 Marks – 2.4 Marks per well-discussed specification):

Hardware Compatibility and System Requirements: Evaluate whether the existing processor architectures, RAM capacity, and storage space meet the minimum thresholds required to run the OS smoothly.

Built-in Security Architecture: Assess the quality of integrated security features, such as built-in firewalls, user access control systems, and automated encryption capabilities to defend against data breaches.

Total Licensing and Maintenance Costs: Balance the initial cost of purchasing software licenses against free, open-source options, factoring in long-term technical support expenses.

User Interface Design and Usability: Evaluate whether the Graphical User Interface (GUI) is easy to use, ensuring staff can operate the system efficiently with minimal retraining.

Application Software Compatibility: Ensure the new OS fully supports the specialized software tools, databases, and drivers required by the institution's workflows.

Availability of System Updates and Patches: Verify that the software vendor provides regular, reliable security updates and system patches to protect against emerging digital threats.

Conclusion (1.5 Marks): Summarize that selecting an operating system should never be based on brand popularity alone. Conclude that a thorough technical review helps IT managers select an OS that ensures system stability, security, and long-term efficiency.

11.

Introduction (1½ marks)

ERP (Enterprise Resource Planning) is an integrated information system that manages and coordinates major organizational activities using a common database.

Main Body (12 marks)

Award 2 marks each, any six.

- Integrates information from all departments.

- Improves decision making through real-time information.
- Reduces duplication of data.
- Improves communication among departments.
- Increases employee productivity.
- Enhances inventory management.
- Improves payroll accuracy.
- Improves customer service.
- Supports better financial management.
- Reduces operating costs.
- Improves report generation.
- Enhances data security through centralized management.

Conclusion (1½ marks)

ERP systems improve organizational efficiency by integrating business processes, reducing costs, improving decision making, and increasing productivity.